

Establishing the diagnosis of Cushing syndrome

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INTRODUCTION

The possible presence of Cushing syndrome is suggested by certain symptoms and signs. Unfortunately, none of these are pathognomonic, and many are nonspecific (eg, obesity, hypertension, menstrual irregularity, and glucose intolerance). As a result, the diagnosis must be confirmed by biochemical tests. The diagnosis of Cushing syndrome involves three steps: suspecting it on the basis of the patient's symptoms and signs, documenting the presence of hypercortisolism, and determining its cause. The evaluation to determine if the patient has hypercortisolism (Cushing syndrome) will be reviewed here. The pathophysiology, causes, clinical manifestations, and approach to the differential diagnosis of established hypercortisolism is discussed separately. (See "[Causes and pathophysiology of Cushing syndrome](#)" and "[Epidemiology and clinical manifestations of Cushing syndrome](#)" and "[Establishing the cause of Cushing syndrome](#)".)

OVERVIEW OF DIAGNOSTIC APPROACH

Who should be tested? — We suggest testing for hypercortisolism in patients in whom a diagnosis is most likely, including the following ([table 1](#)) [1-3]:

- Unusual findings for their age (osteoporosis or hypertension in young adults).

Cushing syndrome: Signs and symptoms	
Hypercortisolemia (Cushing's) and hyperandrogenism	
Hirsutism	
Acne	
Weight gain	
Hypertension	
Hypokalemia	
Glucose intolerance	
Osteoporosis	
Mood changes	
Menstrual irregularities	
Facial rounding	
Dorsal fat pad	
Hypertrophic skin lesions	
Signs and symptoms of hypercortisolemia	
Hypertension	
Hypokalemia	
Glucose intolerance	
Osteoporosis	
Mood changes	
Menstrual irregularities	
Facial rounding	
Dorsal fat pad	
Hypertrophic skin lesions	
Cardiovascular	
Hypertension	
Hypokalemia	
Glucose intolerance	
Osteoporosis	
Mood changes	
Menstrual irregularities	
Facial rounding	
Dorsal fat pad	
Hypertrophic skin lesions	
Neuroendocrine	
Hypertension	
Hypokalemia	
Glucose intolerance	
Osteoporosis	
Mood changes	
Menstrual irregularities	
Facial rounding	
Dorsal fat pad	
Hypertrophic skin lesions	

- Multiple progressive features of [Cushing syndrome](#), particularly those that are predictive of [Cushing syndrome](#) such as facial [plethora](#), proximal [myopathy](#), striae (>1 cm wide and red/purple), and easy [bruising](#).
- Unexplained severe features ([resistant hypertension](#), [osteoporosis](#)) at any age.
- [Adrenal incidentalomas](#).

[Cushing syndrome: Signs and symptoms](#)

Table 1 - larger image below

Exclude exogenous glucocorticoids — Before evaluation for possible [Cushing syndrome](#), it is essential that a careful history has excluded exogenous [glucocorticoid](#) intake, as these patients should not be evaluated for [Cushing syndrome](#):

- The most common cause of [hypercortisolism](#) is ingestion of prescribed [glucocorticoid](#), usually for nonendocrine disease. However, [Cushing syndrome](#) can also be caused by other oral, injected, topical, and intranasal [glucocorticoids](#) [4-6] and by high doses of [megestrol acetate](#) or other [progestins](#) with some intrinsic [glucocorticoid](#) activity [7].

[Ritonavir](#) inhibits CYP3A4 metabolism of many [glucocorticoids](#) and can cause excessive exposure to some inhaled or injected [steroids](#), leading to [Cushing syndrome](#) [8,9]. [Cushing syndrome](#) may also be caused by the use of [glucocorticoid](#)-containing creams or herbal preparations or overuse of [glucocorticoids](#) through any route of administration [10,11].

- All [glucocorticoids](#) (including potent inhaled and topical [glucocorticoids](#), such as [beclomethasone](#) and [flucinolone](#), and especially inhaled [fluticasone](#)) inhibit [corticotropin \(ACTH\)](#) secretion if given in sufficient doses (see "[Major side effects of inhaled glucocorticoids](#)", section on '[Adrenal suppression](#)'). Thus, plasma [ACTH](#) and serum [cortisol](#) concentrations and urinary [cortisol](#) excretion may all be low, unless [hydrocortisone](#) (which is [cortisol](#)) or cortisone is the [steroid](#) administered [12].

Urinary [cortisol](#) excretion may be falsely elevated with vulvovaginal application of [hydrocortisone](#) [13].

surreptitious intake of [glucocorticoids](#) is known as **factitious [Cushing syndrome](#)**, a rare disorder that may be seen in individuals who are close to the health professions [12,14]. Factitious [Cushing syndrome](#) is responsible for less than 1 percent of patients with [Cushing syndrome](#), but even a careful history may fail to detect this disorder and it may be difficult to exclude with laboratory tests [14-16].

Important clues to the diagnosis are low or erratic values for urinary [cortisol](#), suggesting ingestion of a synthetic [glucocorticoid](#) or intermittent ingestion of [hydrocortisone](#) or cortisone [14], or excessive urinary [cortisol](#) values relative to serum [cortisol](#) concentrations, suggesting the addition of [hydrocortisone](#) to urine specimens [16]. The most valuable laboratory test is the detection of synthetic [glucocorticoids](#) in the urine by high-pressure liquid chromatography or by gas chromatography/mass spectrometry (GCMS) [12,14,15]. This approach does not detect excessive ingestion of exogenous [hydrocortisone](#), which is identical to endogenous [cortisol](#).

Initial testing — The initial diagnostic tests for [hypercortisolism](#) should be highly sensitive, even though the diagnosis may be excluded later by more specific tests [17,18]. The diagnosis of [Cushing syndrome](#) is established when at least two different first-line tests are unequivocally abnormal.

Once the diagnosis is established, additional evaluation is done to identify the cause of the [hypercortisolism](#). (See "[Establishing the cause of Cushing syndrome](#)".)

We agree with the diagnostic approach outlined by the evidence-based 2008 Endocrine Society clinical guidelines [1]:

- For patients with a low index of suspicion, we suggest initial testing with one of the following first-line tests: bedtime salivary [cortisol](#) (two measurements), 24-hour urinary free [cortisol](#) (UFC) excretion (two measurements), or the overnight 1 mg [dexamethasone](#) suppression test (DST). For example, a woman with [oligomenorrhea](#) and [hirsutism](#) might be tested for [Cushing syndrome](#); however, the [pretest probability](#) of the syndrome is low if there are no other associated signs or symptoms.
 - For patients with a high index of suspicion, such as those with features suggestive of Cushing syndrome, we do two or three first-line tests (with at least two measurements for UFC or salivary cortisol).
- To optimize sensitivity, we suggest using the upper limit of the reference range for UFC and salivary cortisol and a serum cortisol concentration <1.8 mcg/dL (50 nmol/L) after [dexamethasone](#) as the criteria for a normal response.
- If UFC is chosen as the initial screening test, the result should be unequivocally increased (threefold above the upper limit of normal for the assay), or the diagnosis of Cushing syndrome remains uncertain and other tests should be performed.
 - Some centers use the longer low-dose DST (2 mg/day for 48 hours) as an initial test [1].

- UFC and bedtime salivary cortisol measurements are each obtained at least twice because the hypercortisolism in Cushing syndrome may be variable. Two measurements must be abnormal for the test to be considered abnormal; for patients with mild or fluctuating disease, this may require collecting a number of salivary cortisols or UFCs over weeks.

Normal results — If initial testing is normal in an individual with a low index of suspicion for Cushing syndrome, it is unlikely that the patient has Cushing syndrome unless it is mild or cyclic. We do not suggest additional evaluation unless symptoms progress or cyclic Cushing syndrome is suspected. In this case, we suggest referral to an endocrinologist for repeat testing and further evaluation.

On the other hand, if initial testing is normal in someone with a high index of suspicion for Cushing syndrome (clinical features suggestive of Cushing syndrome), we suggest referral to an endocrinologist for additional evaluation to confirm or exclude the diagnosis of Cushing syndrome.

Any abnormal result — In patients with at least one abnormal test result (which could represent true Cushing syndrome or a false-positive result), we suggest additional evaluation. This includes excluding physiologic hypercortisolism and referral to an endocrinologist. Additional evaluation may include repeating the initial test or other first-line tests.

Exclude physiologic hypercortisolism — Hypercortisolism can occur in several disorders other than Cushing syndrome [2]. When such patients present with clinical features consistent with Cushing syndrome, they may also be referred to as having physiologic hypercortisolism or pseudo-Cushing syndrome. Clinically, patients with these physiologic forms of hypercortisolism seldom have the cutaneous (ie, easy bruising, thinning, and friability) or muscle (ie, proximal muscle atrophy and weakness) signs of Cushing syndrome [17]. These conditions/disorders should be excluded when evaluating patients for Cushing syndrome.

Examples of conditions associated with physiologic hypercortisolism that may have some clinical features of Cushing syndrome include:

- Pregnancy
- Patients with severe obesity, especially those with visceral obesity or polycystic ovary syndrome (PCOS)
- Patients with psychological stress, especially patients with a severe major depressive disorder and melancholic symptoms
- Poorly controlled diabetes mellitus
- Rarely, chronic alcoholism, especially during withdrawal

- Physical stress (illness, hospitalization/surgery, pain)
- Obstructive sleep apnea

Examples of conditions associated with physiologic hypercortisolism that are **unlikely** to have clinical features of Cushing syndrome include:

- Malnutrition, anorexia nervosa
- Intense chronic exercise
- Hypothalamic amenorrhea
- High corticosteroid-binding globulin (CBG) (increased serum cortisol but not UFC)
- Glucocorticoid resistance

We suggest against routine testing for Cushing syndrome in these patients unless they develop features predictive of Cushing syndrome such as wide purplish stria, proximal myopathy, or easy bruising, or if they have multiple brain-related signs and symptoms such as irritability, new or increased depression/anxiety, "brain fog," fatigue, and impaired short-term memory.

The distinction between physiologic hypercortisolism and Cushing syndrome is not always simple because patients with Cushing syndrome can have serious infections, frequently have depression (although it is often an atypical, agitated depression) [19,20], and presumably have a prevalence of chronic alcoholism similar to that of the general population.

The psychiatric literature suggests that as many as 80 percent of patients with major depressive disorders have increased cortisol secretion [21-23]. However, cortisol hypersecretion, when present, is usually mild. Furthermore, even patients with severe depression and substantial cortisol hypersecretion rarely develop clinical Cushing syndrome. However, some patients with depression may be difficult to distinguish clinically or biochemically from those with Cushing disease. Their abnormal cortisol secretion presumably results from hypothalamic-pituitary-adrenal (HPA) axis hyperactivity [22], and disappears after remission of depression [24]. (See "[Unipolar depression: Neurobiology](#)", section on 'Hypothalamic-pituitary-adrenal axis'.)

Approximately 50 patients with chronic alcoholism and clinical or biochemical manifestations of Cushing syndrome have been reported since 1976 [25]. The majority had liver dysfunction, with aspartate aminotransferase greater than alanine aminotransferase, although had liver dysfunction, although the hormonal changes did not correlate closely with the degree of abnormality in liver function. In addition to liver dysfunction, these patients probably have transiently increased secretion of corticotropin-releasing hormone (CRH) or impaired hypothalamic or pituitary responsiveness to cortisol. However, their

peripheral and petrosal sinus plasma CRH concentrations are normal, although these tests are not clinically indicated [26]. The hormonal abnormalities disappear rapidly during abstinence from alcohol.

Females with hypothalamic amenorrhea due to stress or weight loss also may have elevated UFC [27], and male obligate exercisers have increased evening cortisol and ACTH levels [28]. (See "[Functional hypothalamic amenorrhea: Pathophysiology and clinical manifestations](#)".)

UFC and cortisol values normalize in patients with anorexia nervosa who gain weight and normalize body mass index (BMI) [29] (see "[Anorexia nervosa: Endocrine complications and their management](#)", section on 'Adrenal'). This suggests that treatment of other physiologic causes of hypercortisolism may result in normalization of UFC and assist in the discrimination between these physiologic states and Cushing syndrome.

CRH after dexamethasone test — The corticotropin-releasing hormone (CRH; corticorelin) after [dexamethasone](#) test exploits the greater sensitivity of ACTH secretion to dexamethasone suppression in patients with depression or other causes of physiologic hypercortisolism [30,31] and their blunted serum cortisol response to exogenous CRH as compared with healthy people or patients with Cushing disease [23,31]. While this test has diagnostic accuracy similar to other screening tests [32], CRH is no longer available worldwide.

Refer to endocrinologist for additional testing — As noted, in patients with at least one abnormal test result, we suggest referral to an endocrinologist for additional evaluation. Additional evaluation may include repeating the abnormal study or other first-line tests.

Normal (Cushing syndrome unlikely) — We suggest no further testing for Cushing syndrome in patients with negative results on two different tests (unless cyclical disease is suspected, which is rare) or if additional signs and symptoms occur over time.

Abnormal (Cushing syndrome confirmed) — The diagnosis of Cushing syndrome is established when at least two different first-line tests are unequivocally abnormal [1], and physiologic hypercortisolism has been excluded. Once the diagnosis is established, additional evaluation is done to identify the cause of the hypercortisolism. (See "[Establishing the cause of Cushing syndrome](#)".)

For the occasional patient with a high pretest probability of Cushing syndrome but negative or discordant results on initial and repeat testing, we suggest follow-up with additional testing.

Delays in diagnosis — Early diagnosis is important to avoid the complications and excess mortality associated with Cushing syndrome. However, for most patients, it takes years to be diagnosed [33]. In some patients, this may be due to clinical features that are both nonspecific and common in individuals without hypercortisolism (hypertension, obesity, and glucose intolerance). (See "[Epidemiology and clinical manifestations of Cushing syndrome](#)".)

In a meta-analysis of 44 studies (including 5367 patients), the mean time to diagnosis was 38 months (95% CI 33-43) for patients with pituitary Cushing syndrome and 30 months (95% CI 24-36) for adrenal Cushing syndrome. The duration was shorter for those with ectopic Cushing syndrome, likely due to the severity of their hypercortisolism (14 months [95% CI 11-17]) [33].

AVAILABLE TESTS

None of the initial tests have ideal sensitivity or specificity. For example, in one study of 369 patients with overweight or obesity, and two additional features of Cushing syndrome, the specificity of diagnostic tests ranged from 84 to 96 percent [34]. In a meta-analysis of 139 studies in 14,140 participants, all initial screening tests had similar sensitivity and specificity [32]. As no one test is superior to others, the choice of test should be individualized to minimize false-positive results.

For example, a bedtime salivary cortisol test is likely to be abnormal in a shift worker, and a [dexamethasone](#) suppression test (DST) response may be abnormal in a woman taking oral estrogen, because of increased corticosteroid-binding globulin (CBG; and hence total cortisol). These tests would not be ideal, and other tests would be chosen based on the individual's history and lifestyle.

24-hour urinary cortisol excretion — Urinary cortisol excretion over 24 hours provides a direct and reliable practical index of cortisol secretion [35,36]. Corticotropin (ACTH) and cortisol are secreted in discrete bursts, not only in normal subjects but also in most patients with Cushing disease [37-39]; in addition, cortisol is secreted episodically by some adrenal tumors ([figure 1](#))

(. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on [basic physiology of the hypothalamic-pituitary-adrenal axis](#)'.)

24-hour urinary cortisol excretion over 24 hours is an integrated measure of the serum free cortisol concentration (ie, cortisol that is not bound to CBG [transcortin] or other serum proteins). The two most important factors in obtaining a valid result are collection of a complete 24-hour specimen and a reliable reference laboratory. The former can usually be obtained by carefully explaining to the patient how to collect the specimen and by measuring urinary creatinine excretion (see "[Patient education: Collection of a 24-hour urine specimen \(Beyond the Basics\)](#)").

Urinary cortisol assays are reviewed in detail separately. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on 'Assay methods' and "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on 'Caveats for interpretation'.)

The best evidence for the utility of urinary free cortisol (UFC) as a screening test comes from a systematic review and meta-analysis of studies of diagnostic tests for Cushing syndrome performed in conjunction with the 2008 Endocrine Society guidelines [40]. Tests with a high likelihood ratio for a positive (abnormal) result indicate tests that help rule in Cushing syndrome, while those with a very low likelihood ratio for a negative (normal) result indicate tests that help rule it out. Salivary cortisol, UFC, and the overnight DST had similar accuracy. For urinary cortisol, 14 studies were included of 646 patients with Cushing syndrome and 5226 patients who did not have Cushing syndrome. The likelihood ratio of 10.6 (95% CI 5.5-20.5) for an abnormal result and a likelihood ratio for Cushing syndrome of 0.16 (95% CI 0.08-0.33) for a normal result. However, other studies suggest that in cases of mild Cushing syndrome, UFC has less diagnostic sensitivity than salivary cortisol [41,42]. Although liquid chromatography-tandem mass spectrometry (LC-MS/MS) methods are more analytically specific than immunoassays, their use for measuring UFC has been questioned, as they give a lower (and potentially falsely negative) result [43].

Some experts argue that UFC determinations are unnecessary because bedtime serum or salivary cortisol values have similar diagnostic utility and are more convenient (see '[Bedtime salivary cortisol](#)' below). If it is not convenient to collect urine as a second test, then the DST should be used in combination with a bedtime cortisol test.

Interpretation and limitations — The patient can be assumed to have Cushing syndrome if basal urinary cortisol excretion is more than three times the upper limit of normal (which may vary somewhat in different assays) and one other test is abnormal. The patient should then be evaluated for the cause of the hypercortisolism. (See "[Establishing the cause of Cushing syndrome](#)".)

 **Subclinical hypercortisolism** – UFC is of limited utility in patients with subclinical hypercortisolism (eg, patients with mild Cushing syndrome caused by an adrenal incidentaloma). These patients may have normal urine cortisol values but an elevated bedtime salivary cortisol concentration [1]. The overnight DST is suggested for patients with adrenal incidentalomas. (See "[Evaluation and management of the adrenal incidentaloma](#)", section on 'Mild autonomous cortisol secretion'.)

- **False-positive results**

- **Physiologic hypercortisolism** – Patients with equivocally raised urine cortisol values (above normal but less than three times the upper reference value) may have physiologic hypercortisolism (pseudo-Cushing syndrome) [44]. For example, up to 40 percent of patients with severe depression or polycystic ovary syndrome (PCOS) have slightly high 24-hour urinary cortisol excretion [14,45]. In this setting, patients should either be reevaluated after several weeks **or** undergo one or more of the other first-line tests, depending upon the level of clinical suspicion.
- **High fluid intake** – People who drink very large volumes of liquid also excrete more cortisol (64 percent more cortisol excreted with an intake of 5 liters per day), while excretion of creatinine and 17-hydroxycorticosteroids remains unaltered [46]. Therefore, modest increases of urinary cortisol excretion in patients with urine volumes of more than 3 liters should be interpreted with caution.
- **False-negative results** – If urinary cortisol excretion is indeterminate (above the upper limit of the reference range but less than threefold above the upper limit), and the bedtime serum or salivary cortisol concentrations are normal, the patient does not have Cushing syndrome unless it is cyclic or mild [41,47].

A few studies report that some patients with Cushing disease may have normal UFC (measured by tandem mass spectrometry) but are more likely to have abnormal bedtime salivary cortisol [41,48]. In a study of 426 patients with Cushing syndrome, for example, 47 percent of 288 patients with Cushing disease, 31 percent of 80 patients with adrenal adenoma, 21 percent of 25 patients with ectopic ACTH or corticotropin-releasing hormone (CRH) secretion, and 5 percent of 24 patients with adrenal carcinoma had at least one 24-hour urinary cortisol value in this equivocal range [49].

Cortisol excretion decreases progressively with increasing kidney impairment [50]. Therefore, UFC should not be used in this population. (See '[Impaired kidney function](#)' below and "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on '[Caveats for interpretation](#)'.)



These examples illustrate the need for complementary tests, careful consideration of the tests that are chosen, and correlation of the test results.

Bedtime salivary cortisol — A bedtime salivary cortisol concentration can be used to establish the diagnosis of Cushing syndrome [51-58]. Measurement of salivary cortisol in the late evening before bedtime is based upon the fact that the normal evening nadir in serum cortisol is preserved in patients with physiologic hypercortisolism but not in those with Cushing syndrome (see "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)"). Because cortisol

levels appear to be entrained to the initiation of sleep, bedtime salivary cortisol measurement is not a good test for patients with erratic sleep schedules or shift work.

Saliva collection has many advantages: it is noninvasive and can be easily performed by the patient at home, and cortisol is stable in saliva even at room temperature for several days. This test is especially useful for patients suspected of having cyclical or intermittent Cushing syndrome, who can collect many samples over an extended period of time and return the accumulated samples to the laboratory at one time. As with other cortisol assays, it is useful to evaluate at least three samples from different days. Bedtime saliva collection can be performed during 24-hour urine collection for UFC to allow corroboration of test results. (See '[24-hour urinary cortisol excretion](#)' above.)

The criteria used to interpret salivary cortisol results differ among studies because of assay differences. As a result, published reference ranges are not appropriate for all commercial assays [59]. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)".)

In addition, a study of older men (mean age 61 years), some with comorbidities of diabetes and/or hypertension, showed poor test specificity. Twenty percent of the participants had false-positive results (eg, a bedtime salivary cortisol value above the upper limit of normal [1.6 ng/mL, 4.3 nmol/L] when measured by enzyme immunoassay) [60]. None had Cushing syndrome.

In the meta-analysis for the Endocrine Society guidelines described above, bedtime salivary cortisol was an accurate diagnostic test based upon pooled analysis of four studies in 136 patients with Cushing syndrome [40]. The likelihood ratio for an abnormal (positive) result was 8.8 (95% CI 3.5-21.8), and for a normal (negative) result, the likelihood ratio was 0.07 (95% CI 0.00-1.20). The larger meta-analysis that included 1868 patients with Cushing syndrome and 5049 without Cushing syndrome found similar positive (14.6, 95% CI 10.3-20.7) and negative (0.045, 95% CI 0.030-0.066) likelihood ratios [32].

Thus, while a bedtime salivary cortisol measurement is a useful test for the diagnosis of Cushing syndrome, appropriate assay-specific and perhaps age-specific normative values must be used in its interpretation.



In some situations, other second-line tests may be used. These tests may be chosen because of assay-specific constraints, lack of access to salivary cortisol assays, or preferences based on experience.

Bedtime serum cortisol — As noted for bedtime salivary cortisol, measurement of serum cortisol in the late evening, usually at midnight, is also based upon the fact that the normal nadir in serum cortisol after sleep onset is preserved in patients with physiologic hypercortisolism but

not in those with Cushing syndrome. This test is less convenient than the bedtime salivary cortisol and is therefore not used routinely in clinical practice. However, it is sometimes used at specialized centers.

A bedtime serum cortisol >7.5 mcg/dL (207 nmol/L) is used to identify Cushing syndrome. This is based upon a study of 198 patients with Cushing disease, 27 patients with ectopic ACTH syndrome, 15 patients with primary adrenal Cushing syndrome, and 23 patients with pseudo-Cushing syndrome. A single bedtime serum cortisol concentration >7.5 mcg/dL (207 nmol/L) correctly identified 225 of 234 patients with Cushing syndrome; normal values were seen in all 23 patients with pseudo-Cushing syndrome (96 percent sensitivity, 100 percent specificity) [61].

This test has usually been performed in the hospital, and one group obtains the blood after the patient appears to be asleep [62]. However, sleeping versus awake results have not been evaluated systematically, and our experience is that the test can reliably be performed on an ambulatory basis. This can be done by inserting a heparin-lock earlier in the day to avoid stress-induced cortisol release caused by the anticipation of or pain caused by venipuncture and asking the patient to return between 11 PM and midnight for blood drawing. We try to obtain samples on at least two evenings. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)".)

Low-dose dexamethasone suppression tests — The low-dose DSTs are standard **screening** tests to differentiate patients with Cushing syndrome of any cause from patients who do not have Cushing syndrome. The high-dose DSTs are **not** used to make the diagnosis of Cushing syndrome. They are used after the diagnosis of Cushing syndrome is made to distinguish patients with Cushing disease (Cushing syndrome caused by pituitary hypersecretion of ACTH) from patients with ectopic ACTH syndrome (Cushing syndrome caused by nonpituitary ACTH-secreting tumors). (See "[Dexamethasone suppression tests](#)" and "[Establishing the cause of Cushing syndrome](#)".)

There are two forms of low-dose DST: the 1 mg "overnight" and the two-day 2 mg test [19]. These tests are discussed in detail elsewhere. (See "[Dexamethasone suppression tests](#)".)

✦
a DSTs are **not** a good choice for patients in whom CBG levels may be abnormal or in those taking medications that may alter the metabolism of the drug [63]. Estrogen and estrogen-containing drugs, such as oral contraceptives, raise CBG and may result in a false-positive DST. Oral contraceptives should be stopped for six weeks before performing a DST, or an alternate test should be used. For the same reason, DSTs are also not a reliable test in pregnancy. (See '[Pregnancy](#)' below and "[Diagnosis and management of Cushing syndrome during pregnancy](#)".)

- **Overnight 1 mg test** – The overnight test consists of administration of 1 mg of [dexamethasone](#) at 11 PM to 12 AM (midnight) and measurement of serum cortisol at 8 AM the next morning. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)".)

The 2008 Endocrine Society guidelines suggest a diagnostic cortisol criterion of 1.8 mcg/dL (50 nmol/L), recognizing that this choice will optimize sensitivity but decrease specificity. Despite use of this stringent criterion for sensitivity, in one study, 8 percent (6 of 80) patients with Cushing disease showed suppression to less than 2 mcg/dL (55 nmol/L) [64].

The low-dose DST should **not** be used as the sole criterion for the diagnosis of Cushing syndrome. At least one additional test should be done to establish or exclude the diagnosis.

- **Standard two-day 2 mg test** – The two-day 2 mg test, which is done commonly in some countries but not others, consists of administering 0.5 mg of [dexamethasone](#) every six hours for eight doses, and measurement of serum (not urinary) cortisol either two or six hours after the last dose. The same criteria for normal suppression (<1.8 mcg/dL [<50 nmol/L]) used for the 1 mg DST are used for the two-day 2 mg DST. The two-day 2 mg DST is described in greater detail separately. (See "[Dexamethasone suppression tests](#)".)
- **Diagnostic accuracy** – In the meta-analyses described above, the 1 mg DST and the two-day 2 mg DST were both accurate diagnostic tests [32,40] (see '[24-hour urinary cortisol excretion](#)' above):
 - For the 1 mg DST, in 14 studies including 249 patients with Cushing syndrome out of 5305 undergoing testing (using various diagnostic criteria), there was a likelihood ratio of 16.4 (95% CI 9.3-28.8) for an abnormal (positive) result and 0.06 (95% CI 0.03-0.14) for a normal (negative) result [40]. In the larger analysis that included 1531 patients with Cushing syndrome out of 4798 undergoing testing, the likelihood ratios were similar, with a positive likelihood ratio of 10.5 (95% CI 7.2-15.3) and a negative likelihood ratio of 0.016 (95% CI 0.007-0.035) [32].



For the two-day 2 mg test, the first analysis identified eight studies, including 136 patients with Cushing syndrome out of 323 who were tested. It found a likelihood ratio of 7.3 (95% CI 3.6-15.2) for an abnormal result and 0.18 (95% CI 0.06-0.52) for a normal result [40]. The larger analysis included 569 patients with Cushing syndrome out of 1095 undergoing testing. The findings were similar, with a positive likelihood ratio of 13.2 (95% CI 6.47-27.1) and a negative likelihood ratio of 0.051 (95% CI 0.027-0.095) [32].

Serum collected at the time of cortisol measurement should be retained for measurement of [dexamethasone](#) (available in commercial laboratories), to clarify otherwise confusing results

caused by noncompliance and individual variability in, and drug effects on, dexamethasone metabolism [65,66]. (See "[Dexamethasone suppression tests](#)".)

SPECIAL POPULATIONS

Pregnancy — We suggest a 24-hour urinary free cortisol (UFC) test or bedtime salivary cortisol test in pregnant people with suspected Cushing syndrome rather than the low-dose dexamethasone suppression test (DST) [67]. (See "[Diagnosis and management of Cushing syndrome during pregnancy](#)", section on 'Diagnosis'.)

Adrenal incidentaloma — We suggest the 1 mg DST as the first-line test to screen for hypercortisolism in patients with adrenal incidentaloma. Subclinical Cushing syndrome (or mild autonomous cortisol secretion [MACS], cortisol hypersecretion without clinical manifestations of Cushing syndrome) is the most frequent hormonal abnormality detected in patients with adrenal incidentalomas. Although these patients lack many of the usual stigmata of overt Cushing syndrome, they may have one or more of the effects of endogenous cortisol over-secretion (eg, obesity, hypertension, glucose intolerance or diabetes, dyslipidemia, osteoporosis). (See "[Evaluation and management of the adrenal incidentaloma](#)" and '[Interpretation and limitations](#)' above.)

Depending on the amount of cortisol secretion by primary adrenal tumors, the spectrum of biochemical abnormalities will vary. Milder cases will present normal 24-hour UFC levels with slightly elevated bedtime salivary cortisol, incompletely suppressed morning serum cortisol following 1 mg DST, and partially suppressed plasma ACTH levels. In more severe cases, UFC, bedtime cortisol, and serum cortisol following 1 mg DST will be clearly elevated and plasma ACTH may be undetectable [68]. Testing with 1 mg DST is most likely to be abnormal in this population and should be undertaken only if there are metabolic and other features compatible with Cushing syndrome, as patients without any clinical features would not undergo treatment. This topic is reviewed in detail separately. (See "[Evaluation and management of the adrenal incidentaloma](#)".)

 **Cyclic Cushing syndrome** — Some patients have cyclic Cushing syndrome, which is characterized by episodes of cortisol excess alternating with periods of normal cortisol secretion [69]. The episodes of hypercortisolism can occur regularly or irregularly, with intercytic phases ranging from days to months [47]. For patients with suspected cyclic Cushing syndrome, we suggest UFC or bedtime salivary cortisol over the DST. If initial testing is normal but clinical suspicion is high, we suggest follow-up with repeat testing. True cyclic Cushing syndrome is

relatively rare, but variability in UFC excretion is common in all etiologies of Cushing syndrome [70,71].

Impaired kidney function — The validity of UFC measurement requires normal kidney function. For patients with chronic kidney disease, we suggest the 1 mg overnight DST instead of UFC. (See "[Laboratory assessment of hypothalamic-pituitary-adrenal axis function](#)", section on 'Caveats for interpretation'.)

Other — For those with epilepsy, UFC or bedtime cortisols rather than the DST are suggested because antiseizure medications increase [dexamethasone](#) clearance.

The overnight DST is unreliable (false-positive results) in females taking oral contraceptives because of the increase in corticosteroid-binding globulin (CBG). Ideally, oral contraceptives or other forms of estrogen should be held for six weeks before initiating testing.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Diagnosis and treatment of Cushing syndrome](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level

and are best for patients who want in-depth information and are comfortable with some medical terminology.

 These are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Cushing syndrome \(The Basics\)](#)")

- Beyond the Basics topics (see "[Patient education: Cushing syndrome \(Beyond the Basics\)](#)" and "[Patient education: Cushing syndrome treatment \(Beyond the Basics\)](#)")

SUMMARY AND RECOMMENDATIONS

- **Who should be tested?** – We suggest testing for hypercortisolism in patients in whom a diagnosis is most likely, including those with (see '[Who should be tested?](#)' above):
 - Unusual findings for their age (osteoporosis or hypertension in young adults).
 - Multiple progressive features of Cushing syndrome, particularly those that are predictive of Cushing syndrome such as facial plethora, proximal myopathy, striae (>1 cm wide and red/purple), and easy bruising.
 - Unexplained severe features (resistant hypertension, osteoporosis) at any age.
 - Adrenal incidentalomas.
 - **Exclude exogenous glucocorticoid use** – Before evaluation for possible Cushing syndrome, it is essential that a careful history has excluded exogenous glucocorticoid intake as these patients should not be evaluated for Cushing syndrome. (See '[Exclude exogenous glucocorticoids](#)' above.)
 - **Initial testing**
 - For patients with a low index of suspicion, we suggest initial testing with **one** of the following first-line tests: bedtime salivary cortisol (two measurements), 24-hour urinary free cortisol (UFC) excretion (two measurements), or the overnight 1 mg [dexamethasone](#) suppression test (DST). (See '[Initial testing](#)' above.)
 - For patients with a high index of suspicion, we suggest initial testing with **two or three** of the first-line tests: bedtime salivary cortisol (two measurements), 24-hour urinary free cortisol (UFC) excretion (two measurements), or the overnight 1 mg [dexamethasone](#) suppression text (DST). (See '[Initial testing](#)' above.)
- UFC is chosen as the initial screening test, the result should be unequivocally increased (threefold above the upper limit of normal for the assay), or the diagnosis of Cushing syndrome is uncertain and other tests should be performed. (See '[Initial testing](#)' above.)

We suggest using the upper limit of the reference range for UFC and salivary cortisol and a serum cortisol concentration <1.8 mcg/dL (50 nmol/L) after [dexamethasone](#) as the cutoffs for a

normal response. (See ['Initial testing'](#) above.)

Our choice of initial tests in special situations (pregnancy, adrenal incidentalomas, cyclic Cushing syndrome, impaired kidney function) are reviewed above. (See ['Special populations'](#) above.)

- **Additional evaluation** – In patients with at least one abnormal test result (which could represent true Cushing syndrome or a false-positive result), we suggest additional evaluation. This includes excluding physiologic hypercortisolism and referral to an endocrinologist. Additional evaluation may include repeating the initial test or other first-line tests. (See ['Any abnormal result'](#) above.)
- **Establish the diagnosis**
 - The diagnosis of Cushing syndrome is established when at least two different first-line tests are unequivocally abnormal and cannot be explained by other conditions that cause physiologic hypercortisolism. The patient should undergo additional evaluation if only one test is abnormal or only slightly abnormal. (See ['Abnormal \(Cushing syndrome confirmed\)'](#) above.)
 - If test results are normal, the patient does not have Cushing syndrome unless it is extremely mild or cyclic. We do not suggest additional evaluation unless symptoms progress or cyclic Cushing syndrome is suspected. (See ['Normal \(Cushing syndrome unlikely\)'](#) above.)
 - Once the diagnosis is established, additional evaluation is done to identify the cause of the hypercortisolism. (See ["Establishing the cause of Cushing syndrome"](#).)

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GRAPHICS

Table 1: Cushing syndrome: Signs and symptoms

Hyperandrogenism (females) and hypogonadism
Hirsutism
Diminished libido
Oligo- or amenorrhea
<i>Acne</i>
<i>Female balding</i>
Metabolic
Progressive weight gain
Central obesity
Facial rounding
Dorsal fat pad
Impaired glucose tolerance
Skin and musculoskeletal
Ecchymoses
Facial plethora
<i>Striae (>1 cm wide, often violaceous)</i>
<i>Proximal muscle weakness</i>
<i>Back pain</i>
<i>Low bone mineral density</i>
<i>Fragility fracture</i>
Cardiovascular
Hypertension
<i>Peripheral edema</i>
<i>Atherosclerotic cardiovascular disease</i>
Neuropsychiatric
Depressive symptoms or major depression
Emotional lability
Impaired cognition (eg, short-term memory deficits)

<i>Headache</i>
Other
Fatigue
Abdominal pain
<i>Immunocompromise with recurrent infections</i>

Signs and symptoms of Cushing syndrome. *Italicized* features are less common.

ECG: electrocardiogram.

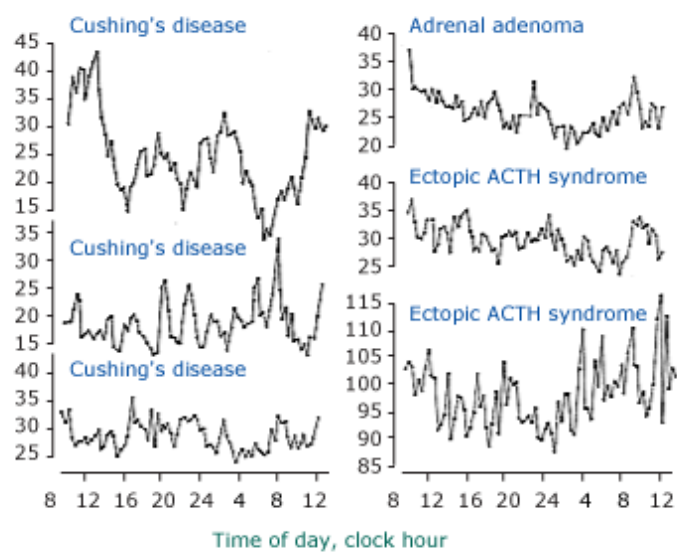
Reference:

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Graphic 111525 Version 10.0



Figure 1: Episodic cortisol secretion in Cushing's syndrome



Serum cortisol concentrations throughout the day in patients with different forms of Cushing's syndrome. There is episodic hormone release in both ACTH-dependent and ACTH-independent (adrenal adenoma) forms of the syndrome. To convert serum cortisol to nmol/L, multiply by 27.6.

ACTH: corticotropin.

Data from: Sederberg-Olsen P, Binder C, Kehlet H, et al. Episodic variation in plasma corticosteroids in subjects with Cushing's syndrome of differing etiology. J Clin Endocrinol Metab 1973; 36:906.

Graphic 60086 Version 5.0



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